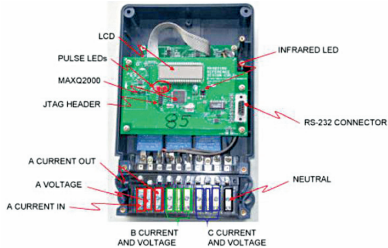
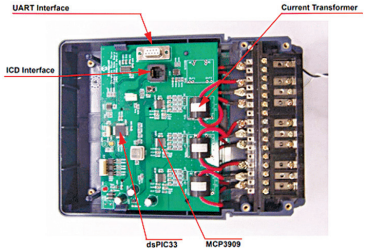


Interesting Reference Designs Of

Title of Reference Design	Precise Polyphase Meter with Temperature Sensing	3-Phase Energy Meter with Harmonics Analysis
Image		
Key Attractions	This reference design for a 3-phase energy meter provides a reliable solution for various metering applications. The design features PC software for easy meter evaluation and calibration. It also offers features such as temperature sensing, phase indication, harmonics measurement, and pulse output for extra functionality.	This reference design for a 3-phase energy meter can provide accurate measurements in a 220V, 5(10)A system. The design provides precise measurements and flexible outputs, suitable for various meter designs, from basic to those needing harmonic analysis.
Highlights	The MAXQ3183 3-phase energy meter reference design from Analog Devices offers accurate energy measurement and power monitoring. The design utilises the polyphase multifunction analogue front end (AFE) and the MAXQ2000 microcontroller. It is designed for 3-phase four-wire systems, operating at AC conditions of 50Hz/60Hz, 220V, 9A. The design offers a meter constant of 6400, achieving a meter accuracy of class 0.5s for active power and class 2 for reactive power measurements. The reference design also includes serial communication with a PC, supporting baud rates of 1200/9600/57600/115200. The design comes with PC software that facilitates easy evaluation and calibration of the meter. This software enables the display and retrieval of various parameters such as voltage root mean square (RMS), current RMS, power factor, harmonics, and more. It also includes calibration screens for adjustments in gain, phase angle, and linearity. The user-friendly graphical interface aids in a more comprehensive understanding and examination of metering functions. Additional features include temperature sensing, phase sequence indicator, harmonics measurement, and pulse output to increase functionality. The inclusion of a serial peripheral interface (SPI) slave interface streamlines communication, and the use of a compact thin shrink small outline package (TSSOP) contributes to space efficiency.	The MCP3909/dsPIC33F reference design from Microchip is tailored for a 3-phase energy meter and is built upon a 16-bit dsPIC33F microcontroller unit (MCU). It is proficient in delivering accurate measurements within a 220V, 5(10)A system. The design employs 16-bit analogue-to-digital converters (ADCs) for precise analogue conversion, while the computational tasks are carried out by a PIC18F2520 flash MCU. Additionally, the design integrates the USB interface module, facilitating meter evaluation and calibration through the '3-phase energy meter software' compatible with Windows XP or Windows 2000. An LCD module is also featured for supplementary energy meter firmware design and evaluation. The reference design can accurately measure active energy pulse output, active power, root mean square (RMS) current and voltage, and over 100 serially accessible output registers. The integration of the dsPIC33F's DSP engine and a quasi-synchronous sampling algorithm leads to cost savings and simplified design. The design provides features like flash-based calculations for streamlined energy measurement, and wide-range ADCs for precise analogue conversion at 128 samples per line cycle. The meter includes calibration and correction registers for precise fine-tuning and user configurability, and with its command-driven operation and mode registers users can customise features like positive-only energy accumulation and no-load threshold mode.
Applications	The reference design is suitable for various metering applications such as electricity metering, water metering, gas metering, etc.	The reference design is suitable for various meter designs, from basic energy meters to those requiring comprehensive harmonic analysis.
OEM Brand	Analog Devices (ADI)	Microchip
URL	https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-for-a-precise-polyphase-meter	https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-for-a-3-phase-energy-meter-with-harmonics-analysis